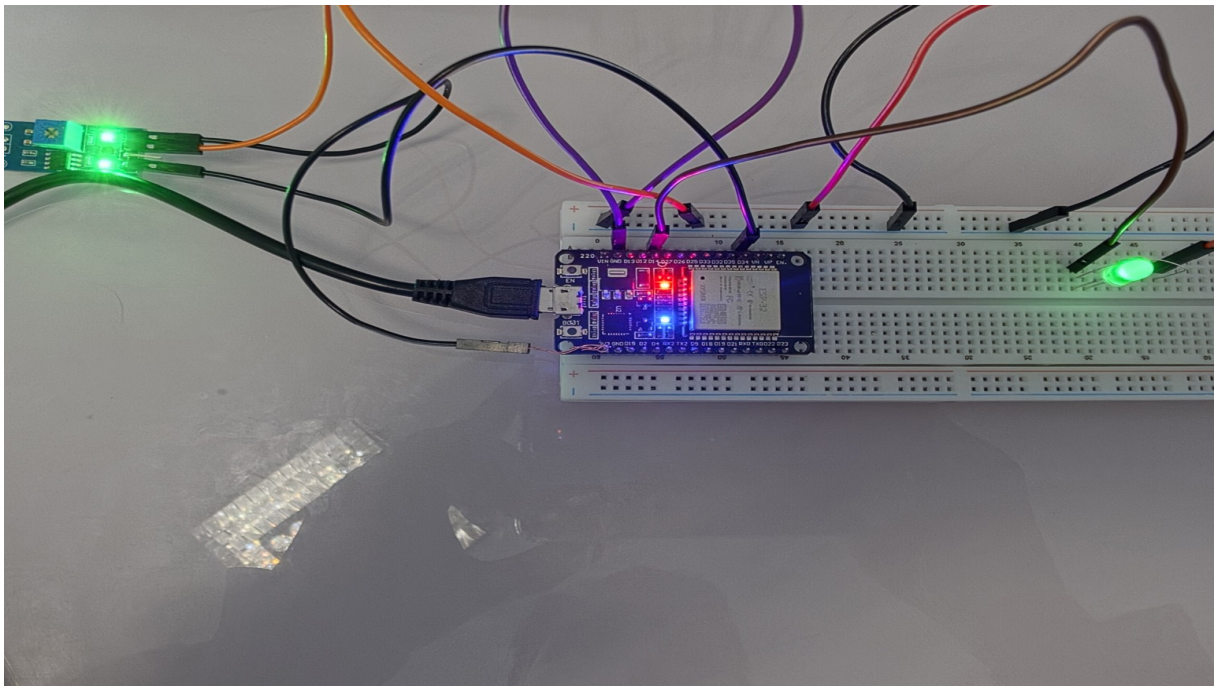
ESP32 Smart Automatic Lighting System

Automatic Light Control using ESP32 & LDR Sensor



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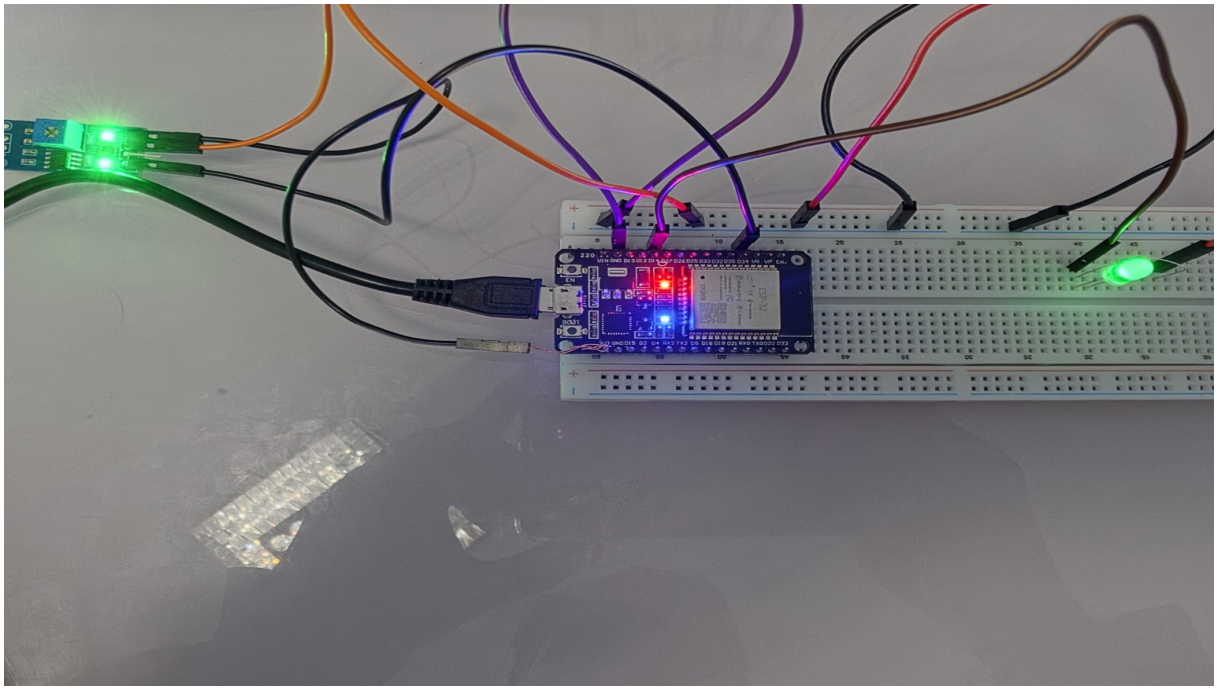
Deutsch

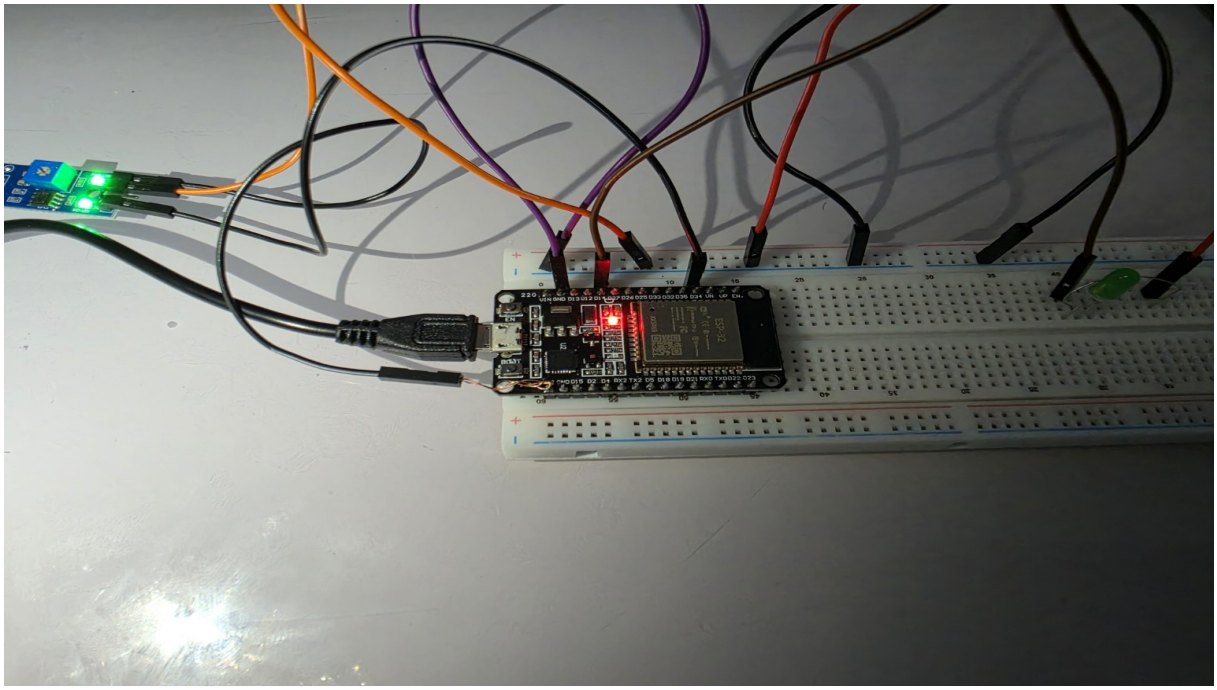
Dieses Projekt demonstriert ein intelligentes automatisches Beleuchtungssystem mit ESP32 und einem LDR-Lichtsensor. Der ESP32 überwacht kontinuierlich die Lichtintensität der Umgebung. Sobald die Umgebung dunkel wird, schaltet das System automatisch die grünen LEDs ein. Bei ausreichendem Licht werden die LEDs automatisch ausgeschaltet. Das Projekt basiert auf MicroPython und zeigt grundlegende Smart-Home- und IoT-Konzepte.

Funktion	Beschreibung
ESP32	WiFi-fähiger Mikrocontroller
LDR Sensor	Erkennung von Licht und Dunkelheit
Automatische LEDs	Automatische Lichtsteuerung
MicroPython	Programmierung des ESP32
GPIO Steuerung	Direkte Hardware-Steuerung
Smart Automation	Automatische Entscheidungslogik

System bei Dunkelheit



System bei Licht



MicroPython Code

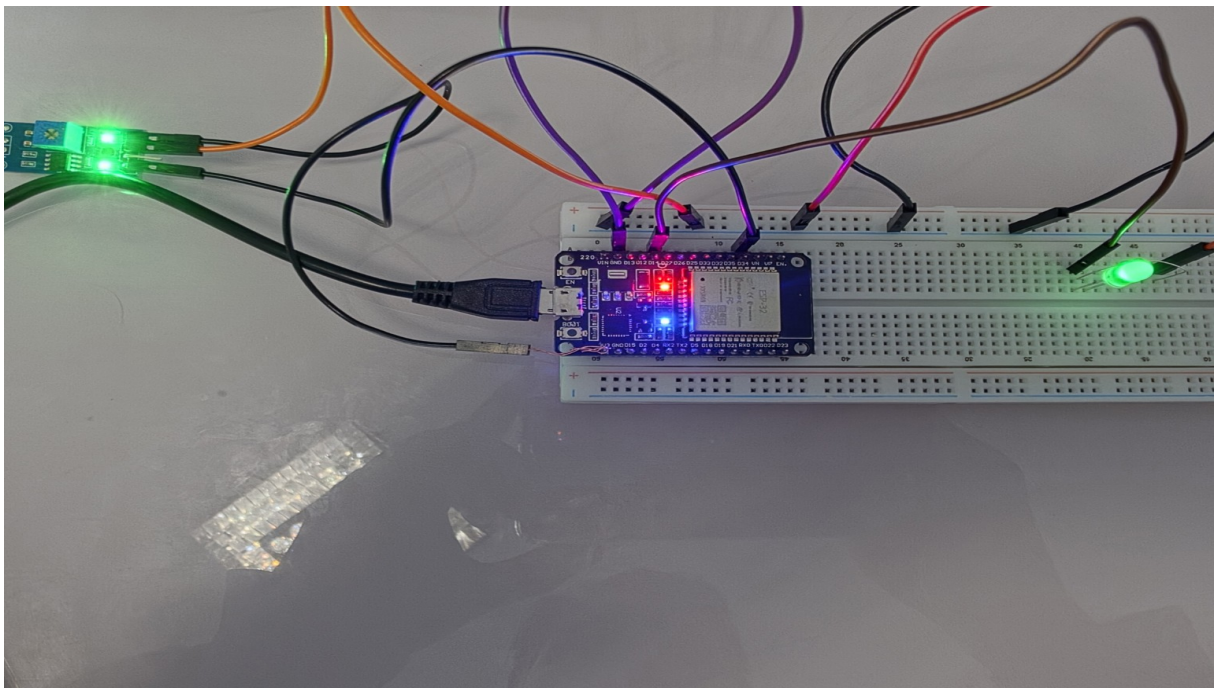
```
from machine import Pin, ADC
import time

green_led = Pin(14, Pin.OUT)
internal_led = Pin(2, Pin.OUT)

light_sensor = ADC(Pin(34))

light_sensor.atten(ADC.ATTN_11DB)
light_sensor.width(ADC.WIDTH_12BIT)

threshold = 1800

while True:

    light_value = light_sensor.read()

    print("Light value:", light_value)

    if light_value > threshold:

        green_led.on()
        internal_led.on()

    else:

        green_led.off()
        internal_led.off()

    time.sleep(0.5)
```

English

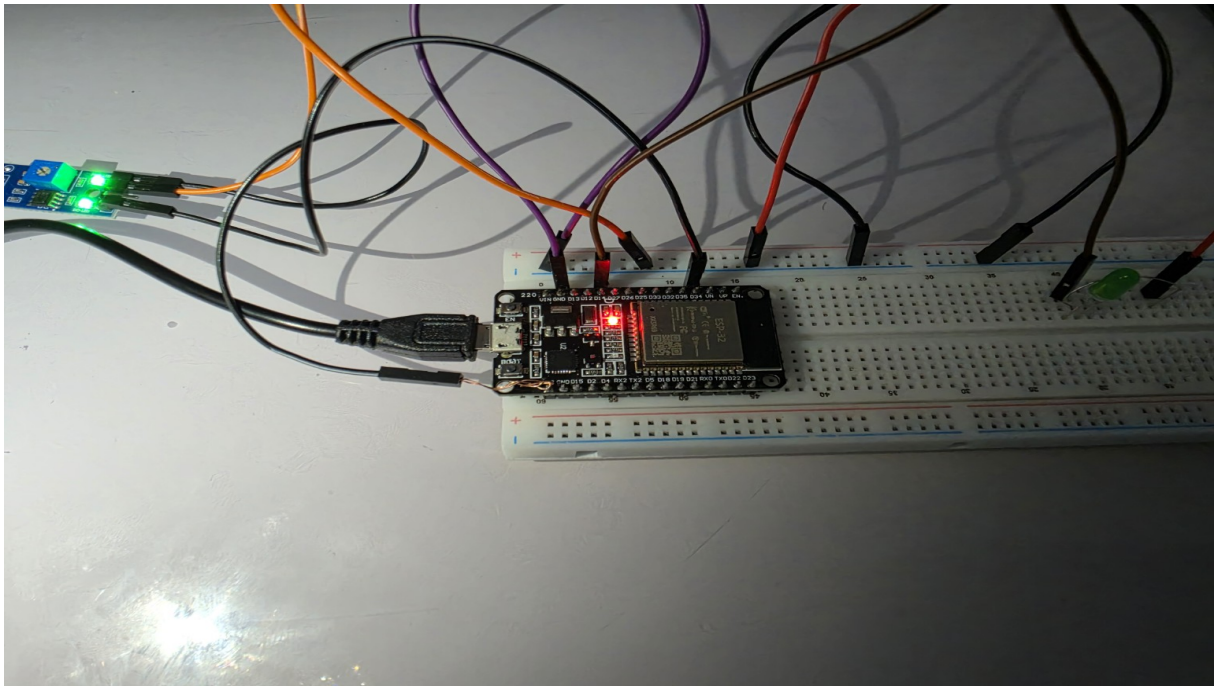
This project demonstrates a smart automatic lighting system using ESP32 and an LDR light sensor. The ESP32 continuously monitors the surrounding light intensity. When the environment becomes dark, the system automatically turns ON the green LEDs. When enough light is detected, the LEDs automatically turn OFF. The project is developed using MicroPython and demonstrates fundamental Smart Home and IoT concepts.

Feature	Description
ESP32	WiFi-enabled microcontroller
LDR Sensor	Light and darkness detection
Automatic LEDs	Automatic light control
MicroPython	ESP32 programming language
GPIO Control	Direct hardware control
Smart Automation	Automatic decision logic

Dark Environment - LEDs ON



Bright Environment - LEDs OFF



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